

Credit Card Portfolio Analysis — Power BI Case Study

End-to-end BI project: SQL data modeling, DAX measures, and multi-page dashboard design

Overview

I built a 3-page Power BI report analyzing a 10,000+ customer credit card portfolio, covering transaction trends, customer segmentation, and credit risk. The project involved designing a relational data model in PostgreSQL, connecting it to Power BI, and building custom DAX measures to surface insights beyond simple revenue totals.

Tools Used

Power BI • PostgreSQL • DAX • Power Query • SQL

Problem

Credit card issuers need visibility not just into total revenue, but into where that revenue comes from, who their customers are, and how much risk sits within the portfolio. This project set out to answer those three questions using real transaction and customer-level data.

Approach

- Designed and loaded a relational database in PostgreSQL, with customer and transaction tables joined on client ID via foreign key constraints
- Connected Power BI directly to the SQL database (Import mode) and built a star-schema model, including a dedicated date table for time intelligence
- Cleaned inconsistent source data in Power Query — fixed casing errors, typos, and mismatched column names across files
- Built custom DAX measures beyond simple aggregations, including a correctly-scoped per-customer acquisition cost measure, a delinquency rate measure, and a utilization-bucketed risk breakdown
- Structured the report into three purpose-built pages: Transaction Report, Customer Report, and Credit Risk & Portfolio Health

- Simulated a real-world data refresh by appending new records directly in SQL and validating that the Power BI report updated correctly end-to-end

Key Insights

1. Acquisition cost doesn't align with revenue.

Platinum cardholders cost the most to acquire per customer (~\$98) but generate the least total revenue of any tier — suggesting acquisition spend may be misallocated relative to the Blue tier, which drives the majority of total revenue.

2. Delinquency risk reverses by age and gender.

Younger male customers (20–30) show nearly double the delinquency rate of younger females in the same age group. This pattern fully reverses by age 60+, where female delinquency becomes the higher of the two.

3. Utilization and delinquency run counter to conventional expectation.

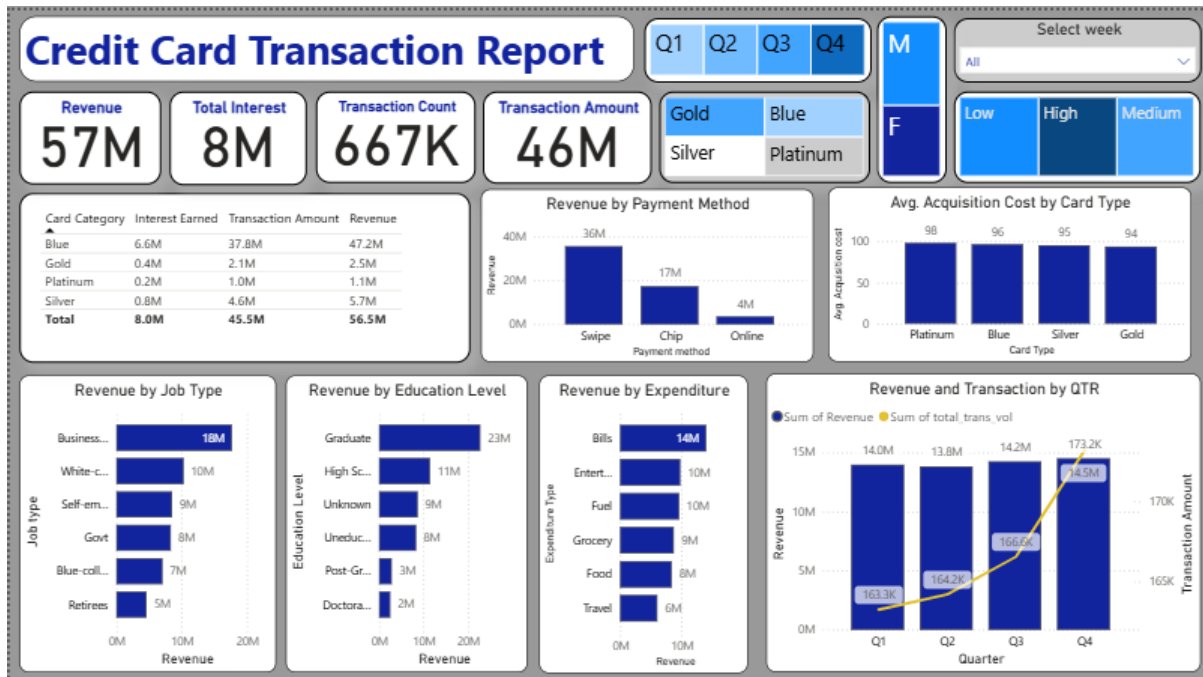
Customers in the lowest utilization bucket showed the highest delinquency rate — the opposite of the typical assumption that higher utilization signals higher risk. This points to a more complex risk profile than utilization alone can explain.

4. Card tier doesn't predict spending behavior.

Blue cardholders carry the lowest credit limit and lowest revolving balance of any tier, yet show by far the highest utilization rate — meaning they operate with the least financial headroom despite being the entry-level card.

Dashboard Pages

Page 1: Transaction Report — revenue, payment behavior, and quarterly trends



Page 2: Customer Report — demographic and behavioral segmentation



Page 3: Credit Risk & Portfolio Health — delinquency, utilization, and credit exposure

